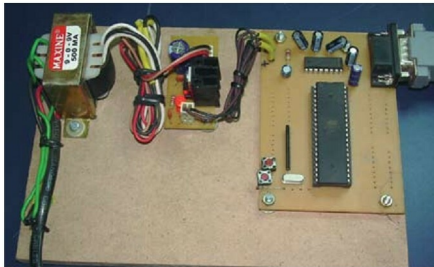


MICROCONTROLLER-BASED SERIAL DATA TRANSFER TO PC

■ T. SHANKAR AND S. THANGAMANI

Communication from or to a computer has become almost essential in our day-to-day lives. Though we can execute many programs within a computer itself, it will not be of much use without outside communication.



Author's prototype

The basic elements of a communication are:

1. A sender (source) that creates the message to be transmitted
2. A medium that carries the message
3. A receiver (sink) that receives the message

There are two types of communications—parallel and serial. Parallel data transfer can be made through an 8-bit data line for faster communication.

Serial communication is often used to control or receive data from a microcontroller. It is a form of input/output (I/O) in which the bits of a byte being transferred appear one after the other in a timed sequence on a single wire. Serial data transfer is made through a

serial link between the microcontroller and the PC using RS-232C.

RS-232C

Fig. 1 shows the waveform on a single conductor to transmit a byte (say, 0x41) serially. The upper waveform is the TTL-level waveform at the transmit pin of the microcontroller (AT89552).



bit. Error-free serial communication requires the baud rate, number of data bits, number of stop bits, and presence or absence of a parity bit to be the same at the transmitter and receiver ends.

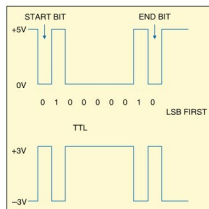


Fig. 1: Serial waveform

PARTS LIST

Semiconductors:

- IC1 - AT89552 microcontroller
- IC2 - MAX232 RS-232 driver
- IC3 - 7805, 5V regulator
- D1-D4 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

- R1 - 330-ohm
- R2 - 10-kilo-ohm
- RNW1 - 10-kilo-ohm resistor network

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2, C3 - 0.1 μ F ceramic disk
- C4, C5 - 22pF ceramic disk
- C6-C11 - 10 μ F, 16V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 300mA secondary transformer
- S1-S9 - Push-to-on switch
- S10 - On/off switch
- X_{TAL} - 11.0592MHz crystal
- 9-pin D-type male/female COM port connector

The lower waveform is the same waveform converted by the RS-232 driver (MAX232) into RS-232C level. The voltage level of the RS-232C is used to assure error-free transmission over greater distances than would be possible with TTL levels.

Each byte is preceded by a start bit and followed by a stop bit. The start and stop bits are used to synchronise the serial receivers. The data byte is always transmitted with least-significant bit (LSB) first. For error checking, it is possible to include a parity bit as well, just prior to the stop bit.

The bits are transmitted at specific time intervals determined by the baud rate of the serial signal. The baud rate is the reciprocal of the time to send one